Tugas Besar Dasar Pemrograman

Asisten Laboratorium 1: Ferdinand Macalister



Disusun Oleh:

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Kom	: A
Laboratorium	1

UNIVERSITAS SUMATERA UTARA
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SOAL 1

onlinegdb.com/?des

main.pas | Soal1_AbiNubliRidh...

```pascal
//Abi Nubli Ridhoi 261401056 Kom A


program tubes1;
uses crt;
var
 n, i :Integer;
 harga, totalharga, total, diskon: real;

begin

//nilai total dan inputan

 total:=0;
 Write('Masukkan jumlah barang yang dibeli '); readln(n);

//pengulangan

   for i:=1 to n do
    begin
      write('barang ke-', i); writeln;
      write('Masukkan harga barang : '); readln(harga);
      total:=total + harga;
    end;

//kondisi (diskon)

   if total < 100000 then
    begin
      diskon:=0
    end
   else if total < 500000 then
    begin
      diskon:=total * 0.1
    end
   else
    begin
      diskon:=total * 0.2;
    end;

//nilai total harga, dan output

  totalharga:=total - diskon;

  writeln;
  writeln('RINCIAN BELANJA');
  writeln('Jumlah barang yang dibeli : ', n);
  writeln('Total Harga (sebelum diskon) ', total:0:2) ;
  writeln('Diskon : ', diskon :0:2);
  writeln('Total Harga : ', totalharga:0:2) ;

end.

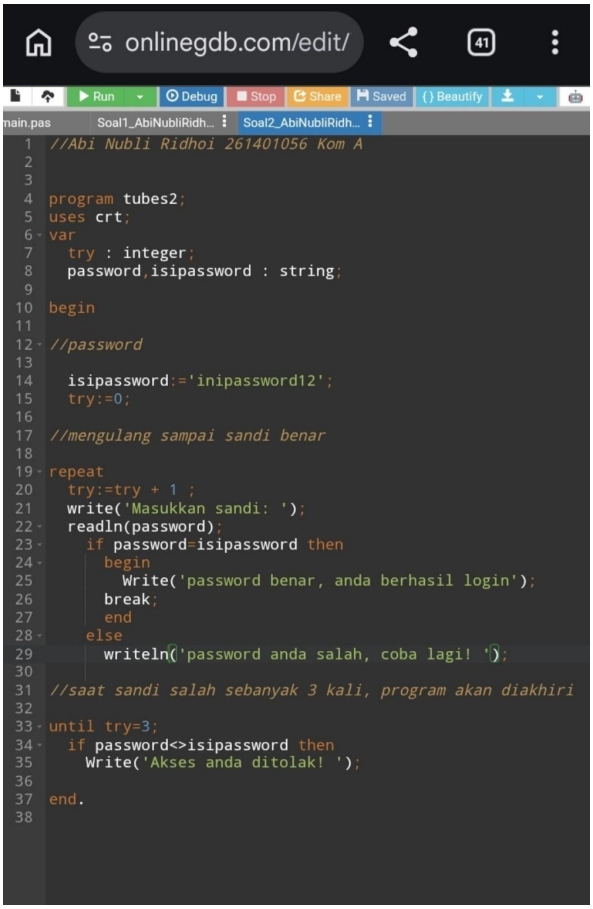
```

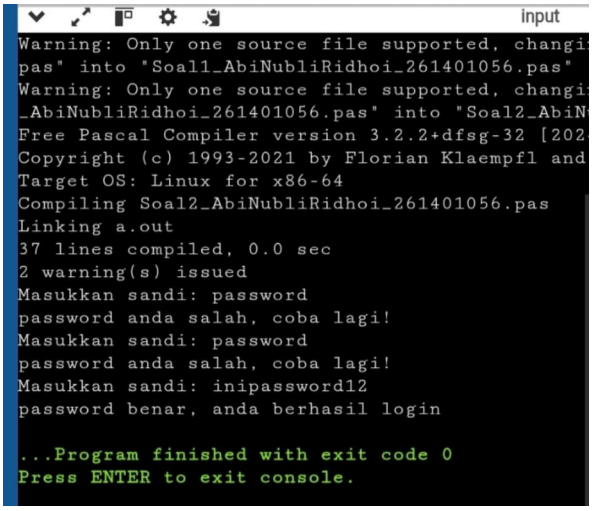
```
Warning: Only one source file supported, changing sour
pas" into "Soal1_AbiNubliRidhoi_261401056.pas"
Free Pascal Compiler version 3.2.2+dfsg-32 [2024/01/05
Copyright (c) 1993-2021 by Florian Klaempfl and others
Target OS: Linux for x86-64
Compiling Soal1_AbiNubliRidhoi_261401056.pas
Linking a.out
52 lines compiled, 0.1 sec
1 warning(s) issued
Masukkan jumlah barang yang dibeli 3
barang ke-1
Masukkan harga barang : 100000
barang ke-2
Masukkan harga barang : 50000
barang ke-3
Masukkan harga barang : 75000

RINCIAN BELANJA
Jumlah barang yang dibeli : 3
Total Harga (sebelum diskon) 225000.00
Diskon : 22500.00
Total Harga : 202500.00


...Program finished with exit code 0
Press ENTER to exit console.
```

SOAL 2


onlinegdb.com/edit/
Run
Debug
Stop
Share
Saved
{ } Beautify
main.pas
Soal1_AbiNubliRidh...
Soal2_AbiNubliRidh...
//Abi Nubli Ridhoi 261401056 Kom A
program tubes2;
uses crt;
var
try : integer;
password,isipassword : string;
begin
//password
isipassword:='inipassword12';
try:=0;
//mengulang sampai sandi benar
repeat
try:=try + 1 ;
write('Masukkan sandi: ');
readln(password);
if password=isipassword then
begin
Write('password benar, anda berhasil login');
break;
end
else
writeln('password anda salah, coba lagi! ');
//saat sandi salah sebanyak 3 kali, program akan diakhiri
until try=3;
if password<>isipassword then
Write('Akses anda ditolak! ');
end.


input
Warning: Only one source file supported, changi
pas" into "Soal1_AbiNubliRidhoi_261401056.pas"
Warning: Only one source file supported, changi
_AbiNubliRidhoi_261401056.pas" into "Soal2_AbiN
Free Pascal Compiler version 3.2.2+dfsg-32 [202
Copyright (c) 1993-2021 by Florian Klaempfl and
Target OS: Linux for x86-64
Compiling Soal2_AbiNubliRidhoi_261401056.pas
Linking a.out
37 lines compiled, 0.0 sec
2 warning(s) issued
Masukkan sandi: password
password anda salah, coba lagi!
Masukkan sandi: password
password anda salah, coba lagi!
Masukkan sandi: inipassword12
password benar, anda berhasil login
...Program finished with exit code 0
Press ENTER to exit console.

SOAL 3

onlinegdb.com/edit/

Run | Debug | Stop | Share | Saved | {} Beautify | Language: Pasc

Soal1_AbiNubliRidh... | Soal2_AbiNubliRidh... | Soal3_AbiNubliRidh...

```
//Abi Nubli Ridhoi 261401056 Kom A


program tubes3;
uses crt;
var
  i, n, choose:integer;

begin

//input

  Write('Masukkan jumlah bilangan ');readln(n);
  WriteLn('pilih kategori');
  WriteLn('pilihan 1 (ganjil)');
  WriteLn('pilihan 2 (genap)') ;
  Write('pilihan ');
  ReadLn(choose);

//menghitung deret berdasarkan ganjil dan melompati kelipatam 5
  i:=1;

   while i<=n do
     begin
       if i mod 5 = 0 then
         begin
           i:=i+1;

         continue;
         end

       else if (choose=1) and (i mod 2 =0) then
         begin
           i:=i+1;
         continue;
         end

       else if (choose =2) and (i mod 2 <> 0) then
         begin
           i:=i+1;
         continue;
         end;

//output

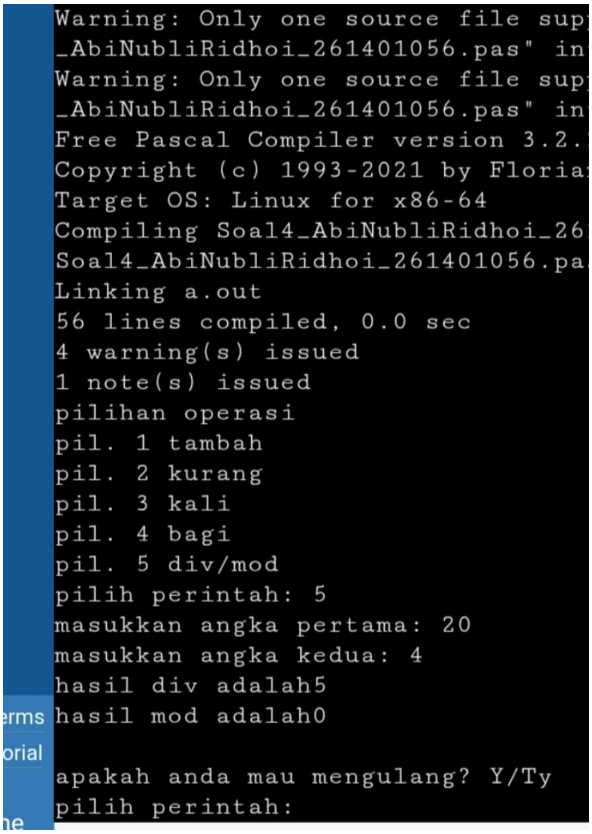
    write(i, ' ');
    i:=i+1;
    end;

  readln;
end.
```

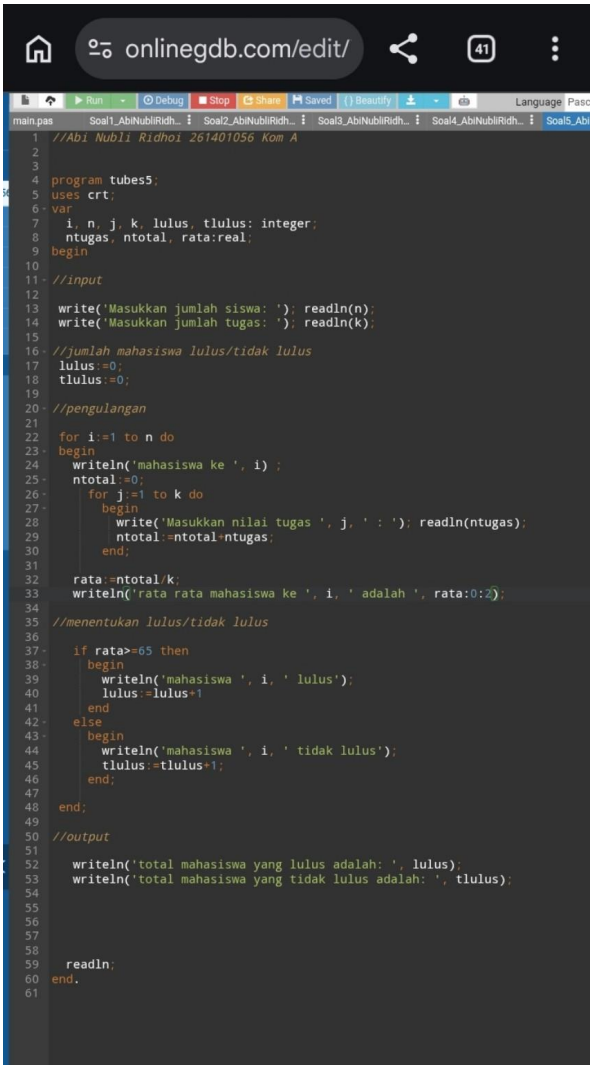
```
Warning: Only one source file supported, c
pas" into "Soal1_AbiNubliRidhoi_261401056.
Warning: Only one source file supported, c
_AbiNubliRidhoi_261401056.pas" into "Soal2
Warning: Only one source file supported, c
_AbiNubliRidhoi_261401056.pas" into "Soal3
Free Pascal Compiler version 3.2.2+dfsg-32
Copyright (c) 1993-2021 by Florian Klaempf
Target OS: Linux for x86-64
Compiling Soal3_AbiNubliRidhoi_261401056.p
Linking a.out
51 lines compiled, 0.0 sec
3 warning(s) issued
Masukkan jumlah bilangan 20
pilih kategori
pilihan 1 (ganjil)
pilihan 2 (genap)
pilihan 2
2 4 6 8 12 14 16 18
```

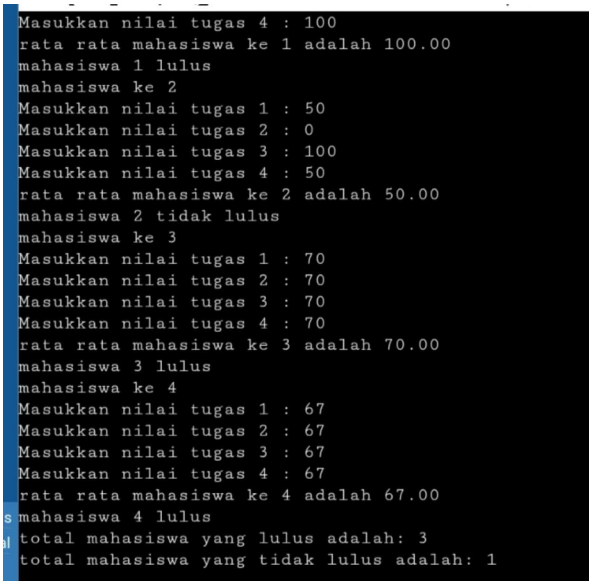
SOAL 4


onlinegdb.com/edit/
Run
Debug
Stop
Share
Saved
Beautify
Soal1_AbiNubliRidh...
Soal2_AbiNubliRidh...
Soal3_AbiNubliRidh...
Soal4_AbiNubliRi
1 //Abi Nubli Ridhoi 261401056 Kom A
2
3
4 program tubes4;
5 uses crt;
6 var
7 operasi, ang1, ang2, i : integer;
8 ulang: char;
9 begin
10
11 //input
12
13 WriteLn('pilihan operasi');
14 WriteLn('pil. 1 tambah');
15 WriteLn('pil. 2 kurang');
16 WriteLn('pil. 3 kali');
17 WriteLn('pil. 4 bagi');
18 WriteLn('pil. 5 div/mod');
19
20 //menentukan pilihan
21 //menentukan pengulangan program menggunakan repeat until
22
23 repeat
24 write('pilih perintah: ');readln(operasi);
25 if operasi>5 then
26 begin
27 WriteLn('tidak ada operasi tersebut');
28 exit;
29 end;
30 Write('masukkan angka pertama: '); readln(ang1);
31 Write('masukkan angka kedua: ');readln(ang2);
32
33
34 case operasi of
35 1:writeln('hasil tambah adalah: ', ang1 + ang2);
36 2:writeln('hasil kurang adalah: ', ang1 - ang2);
37 3:writeln('hasil kali adalah: ', ang1 * ang2);
38 4:writeln('hasil tambah adalah: ', ang1 / ang2);
39
40 //operasi 5 (div dan mod)
41
42 else if operasi=5 then
43 begin
44 writeln('hasil div adalah', ang1 div ang2);
45 writeln('hasil mod adalah', ang1 mod ang2);
46 end;
47 end;
48 writeln;
49 write('apakah anda mau mengulang? Y/T');readln(ulang);
50
51 //pengulangan program sampai pengguna menginput t atau T
52
53 until (ulang='t') or (ulang='T');
54 writeln('program selesai');
55
56 end.
57


Warning: Only one source file sup
_AbiNubliRidhoi_261401056.pas" in
Warning: Only one source file sup
_AbiNubliRidhoi_261401056.pas" in
Free Pascal Compiler version 3.2.
Copyright (c) 1993-2021 by Floria
Target OS: Linux for x86-64
Compiling Soal4_AbiNubliRidhoi_26
Soal4_AbiNubliRidhoi_261401056.pa
Linking a.out
56 lines compiled, 0.0 sec
4 warning(s) issued
1 note(s) issued
pilihan operasi
pil. 1 tambah
pil. 2 kurang
pil. 3 kali
pil. 4 bagi
pil. 5 div/mod
pilih perintah: 5
masukkan angka pertama: 20
masukkan angka kedua: 4
hasil div adalah5
hasil mod adalah0
apakah anda mau mengulang? Y/Ty
pilih perintah:

SOAL 5


onlinegdb.com/edit/
Run
Debug
Stop
Share
Saved
Beautify
Language
main.pas
Soal1_AbiNubliRidh...
Soal2_AbiNubliRidh...
Soal3_AbiNubliRidh...
Soal4_AbiNubliRidh...
Soal5_Abi
//Abi Nubli Ridhoi 261401056 Kom A
program tubes5;
uses crt;
var
i, n, j, k, lulus, tlulus: integer;
ntugas, ntotal, rata:real;
begin
//input
write('Masukkan jumlah siswa: '); readln(n);
write('Masukkan jumlah tugas: '); readln(k);
//jumlah mahasiswa lulus/tidak lulus
lulus:=0;
tlulus:=0;
//pengulangan
for i:=1 to n do
begin
writeln('mahasiswa ke ', i) ;
ntotal:=0;
for j:=1 to k do
begin
write('Masukkan nilai tugas ', j, ' : '); readln(ntugas);
ntotal:=ntotal+ntugas;
end;
rata:=ntotal/k;
writeln('rata rata mahasiswa ke ', i, ' adalah ', rata:0:2);
//menentukan lulus/tidak lulus
if rata>=65 then
begin
writeln('mahasiswa ', i, ' lulus');
lulus:=lulus+1
end
else
begin
writeln('mahasiswa ', i, ' tidak lulus');
tlulus:=tlulus+1;
end;
end;
//output
writeln('total mahasiswa yang lulus adalah: ', lulus);
writeln('total mahasiswa yang tidak lulus adalah: ', tlulus);
readln;
end.


Masukkan nilai tugas 4 : 100
rata rata mahasiswa ke 1 adalah 100.00
mahasiswa 1 lulus
mahasiswa ke 2
Masukkan nilai tugas 1 : 50
Masukkan nilai tugas 2 : 0
Masukkan nilai tugas 3 : 100
Masukkan nilai tugas 4 : 50
rata rata mahasiswa ke 2 adalah 50.00
mahasiswa 2 tidak lulus
mahasiswa ke 3
Masukkan nilai tugas 1 : 70
Masukkan nilai tugas 2 : 70
Masukkan nilai tugas 3 : 70
Masukkan nilai tugas 4 : 70
rata rata mahasiswa ke 3 adalah 70.00
mahasiswa 3 lulus
mahasiswa ke 4
Masukkan nilai tugas 1 : 67
Masukkan nilai tugas 2 : 67
Masukkan nilai tugas 3 : 67
Masukkan nilai tugas 4 : 67
rata rata mahasiswa ke 4 adalah 67.00
mahasiswa 4 lulus
total mahasiswa yang lulus adalah: 3
total mahasiswa yang tidak lulus adalah: 1

SOAL 6

onlinegdb.com/edit/

main.pas | Soal1_AbiNubliRidh... | Soal2_AbiNubliRidh... | Soal3_AbiNubliRidh... | Soal4_AbiNubliRidh...

```
//Abi Nubli Ridhoi 261401056 Kom A


program tubes6;
uses crt;
var
  nakhir, tugas, uts, uas, kehadiran, kehadiran2:real;

begin

//Input

  Write('Masukkan nilai tugas      : ');readln(tugas);
  Write('Masukkan nilai uts        : ');readln(uts);
  Write('Masukkan nilai uas        : ');readln(uas);
  Write('Masukkan jumlah kehadiran : ');readln(kehadiran);

//nilai akhir

  nakhir:=(tugas*0.3)+(uts*0.3)+(uas*0.4);

//penentuan indeks dengan percabangan

if nakhir >=85 then
  begin
    writeln('nilai anda: ', nakhir:0:2, 'dengan ideks A');
  end

else if (nakhir >=75) and (nakhir<=84) then
  begin
    writeln('nilai anda: ', nakhir:0:2, 'dengan ideks B');
  end

else if (nakhir>=60) and (nakhir<=74) then
  begin
writeln('nilai anda: ', nakhir:0:2, 'dengan ideks C');
end

else if (nakhir>=50) and (nakhir<=59) then
  begin
    writeln('nilai anda: ', nakhir:0:2, 'dengan ideks D');
  end
else
  begin
    writeln('nilai anda: ', nakhir:0:2, 'dengan ideks E');
  end;

  writeln;
  writeln('nilai kehadiran anda adalah: ', kehadiran*10:0:2);

//output penentuan lulus/tidak dengan percabangan

if (nakhir>=60) and (kehadiran/10*100>=80) then
  begin
    writeln('Anda lulus! ');
  end

else
  begin
    writeln('Anda tidak lulus! ');
  end;

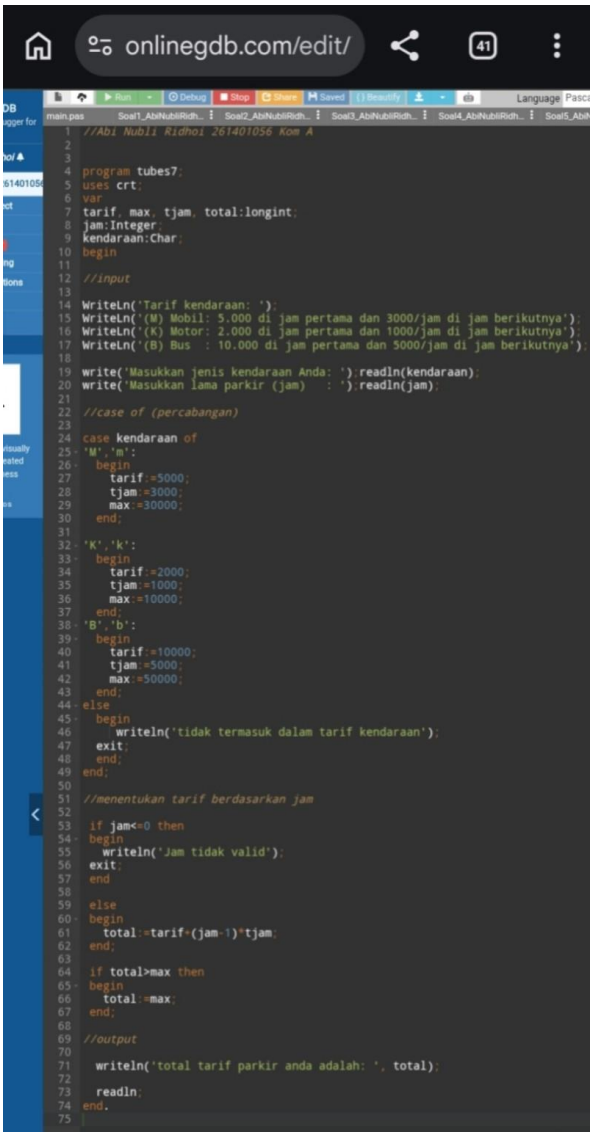

  readln;

end.
```

```
_AbiNubliRidhoi_261401056.pas" into "Soal2_AbiNubl
Warning: Only one source file supported, changing
_AbiNubliRidhoi_261401056.pas" into "Soal3_AbiNubl
Warning: Only one source file supported, changing
_AbiNubliRidhoi_261401056.pas" into "Soal4_AbiNubl
Warning: Only one source file supported, changing
_AbiNubliRidhoi_261401056.pas" into "Soal5_AbiNubl
Warning: Only one source file supported, changing
_AbiNubliRidhoi_261401056.pas" into "Soal6_AbiNubl
Free Pascal Compiler version 3.2.2+dfsg-32 [2024/0
Copyright (c) 1993-2021 by Florian Klaempfl and ot
Target OS: Linux for x86-64
Compiling Soal6_AbiNubliRidhoi_261401056.pas
Soal6_AbiNubliRidhoi_261401056.pas(7,39) Note: Loc
Linking a.out
66 lines compiled, 0.1 sec
6 warning(s) issued
1 note(s) issued
Masukkan nilai tugas      : 100
Masukkan nilai uts        : 100
Masukkan nilai uas        : 100
Masukkan jumlah kehadiran : 10
nilai anda: 100.00dengan ideks A

nilai kehadiran anda adalah: 100.00
Anda lulus!
```

SOAL 7



```
//Abi Nubli Ridhoi 261401056 Kom A


program tubes7;
uses crt;
var
tarif, max, tjam, total:longint;
jam:Integer;
kendaraan:Char;
begin

//input

WriteLn('Tarif kendaraan: ');
WriteLn('(M) Mobil: 5.000 di jam pertama dan 3000/jam di jam berikutnya');
WriteLn('(K) Motor: 2.000 di jam pertama dan 1000/jam di jam berikutnya');
WriteLn('(B) Bus  : 10.000 di jam pertama dan 5000/jam di jam berikutnya');

write('Masukkan jenis kendaraan Anda: ');readln(kendaraan);
write('Masukkan lama parkir (jam)   : ');readln(jam);

//case of (percabangan)

case kendaraan of
'M','m':
  begin
    tarif:=5000;
    tjam:=3000;
    max:=30000;
  end;

'K','k':
  begin
    tarif:=2000;
    tjam:=1000;
    max:=10000;
  end;
'B','b':
  begin
    tarif:=10000;
    tjam:=5000;
    max:=50000;
  end;
else
  begin
    writeln('tidak termasuk dalam tarif kendaraan');
  exit;
  end;
end;

//menentukan tarif berdasarkan jam

 if jam<=0 then
 begin
   writeln('Jam tidak valid');
 exit;
 end

 else
 begin
   total:=tarif+(jam-1)*tjam;
 end;

 if total>max then
 begin
   total:=max;
 end;

//output

  writeln('total tarif parkir anda adalah: ', total);

  readln;
end.
```



```
input
Warning: Only one source file supported, changi
_AbiNubliRidhoi_261401056.pas" into "Soal2_AbiN
Warning: Only one source file supported, changi
_AbiNubliRidhoi_261401056.pas" into "Soal3_AbiN
Warning: Only one source file supported, changi
_AbiNubliRidhoi_261401056.pas" into "Soal4_AbiN
Warning: Only one source file supported, changi
_AbiNubliRidhoi_261401056.pas" into "Soal5_AbiN
Warning: Only one source file supported, changi
_AbiNubliRidhoi_261401056.pas" into "Soal6_AbiN
Warning: Only one source file supported, changi
_AbiNubliRidhoi_261401056.pas" into "Soal7_AbiN
Free Pascal Compiler version 3.2.2+dfsg-32 [202
Copyright (c) 1993-2021 by Florian Klaempfl and
Target OS: Linux for x86-64
Compiling Soal7_AbiNubliRidhoi_261401056.pas
Linking a.out
74 lines compiled, 0.0 sec
7 warning(s) issued
Tarif kendaraan:
(M) Mobil: 5.000 di jam pertama dan 3000/jam di
(K) Motor: 2.000 di jam pertama dan 1000/jam di
(B) Bus  : 10.000 di jam pertama dan 5000/jam di
Masukkan jenis kendaraan Anda: m
Masukkan lama parkir (jam)   : 45
total tarif parkir anda adalah: 30000
```

SOAL 8

onlinegdb.com/edit/

```pascal
//Abi Nubli Ridhoi 261401056 Kom A


program tubes8;
uses crt;
var
jam, lembur:integer;
golongan:char;
bonus, gpokok, gtotal:longint;
begin

//Inputan

Write('Masukkan golongan: ');readln(golongan);
write('Masukkan total jam kerja: ');readln(jam);

//Case of (percabangan)

case golongan of
'A','a':
  begin
  gpokok:=1500000;
  end;

'B','b':
  begin
  gpokok:=2000000;
  end;

'C','c':
  begin
  gpokok:=2500000;
  end;


else
begin
writeln('tidak ada golongan tersebut');
exit;
end;
end;

//penentuan lembur dan bonus

 if jam<40 then
 begin
 writeln('Jam kerja tidak mencukupi untuk mendapat gaji');
 exit;
 end

 else
 begin
 gtotal:=gpokok + (jam-40) * 20000;
 end;

//case khusus golongan c

 if ((golongan='C') or (golongan='c')) and (jam>50) then
 begin
 gtotal:=gpokok + 100000 + (jam-40) * 20000;
 end;

//output

writeln('gaji pokok anda adalah ', gpokok);
WriteLn('jumlah jam lembur anda adalah ', jam-40);
writeln('jumlah bonus anda adalah ', gtotal-gpokok);
writeln('jumlah total gaji anda adalah ', gtotal);

  readln;
end.
```

```
Warning: Only one source file supported, ch
_AbiNubliRidhoi_261401056.pas" into "Soal14_
Warning: Only one source file supported, ch
_AbiNubliRidhoi_261401056.pas" into "Soal15_
Warning: Only one source file supported, ch
_AbiNubliRidhoi_261401056.pas" into "Soal16_
Warning: Only one source file supported, ch
_AbiNubliRidhoi_261401056.pas" into "Soal17_
Warning: Only one source file supported, ch
_AbiNubliRidhoi_261401056.pas" into "Soal18_
Free Pascal Compiler version 3.2.2+dfsg-32
Copyright (c) 1993-2021 by Florian Klaempfl
Target OS: Linux for x86-64
Compiling Soal8_AbiNubliRidhoi_261401056.pa
Soal8_AbiNubliRidhoi_261401056.pas(7,6) Not
Soal8_AbiNubliRidhoi_261401056.pas(9,1) Not
Linking a.out
71 lines compiled, 0.0 sec
8 warning(s) issued
2 note(s) issued
Masukkan golongan: c
Masukkan total jam kerja: 56
gaji pokok anda adalah 2500000
jumlah jam lembur anda adalah 16
jumlah bonus anda adalah 420000
jumlah total gaji anda adalah 2920000
```

• Terms
Tutorial

SOAL 9

onlinegdb.com/edit/

```
//Abi Nubli Ridhoi 261401056 Kom A


program tubes9;
uses crt;
var
bulan, tahun, hari, kabisat:integer;

//Inputan

begin
  Write('Masukkan tahun: ');readln(tahun);
  Write('Masukkan bulan: ');readln(bulan);

//Percabangan

  case bulan of
  1,3,5,7,8,10,12:hari:=31;
  4,6,9,11:hari:=30;

  else if bulan>12 then
  begin
  writeln('Tidak ada bulan tersebut');
  exit;
  end;
  end;

//menentukan tahun kabisat

  if ((tahun mod 4=0) and (tahun mod 100 <> 0)) or (tahun mod 400=0) then
  begin
  kabisat:=1;
  end
  else
  begin
  kabisat:=0;
  end;

  if (kabisat=1) and (bulan=2) then
  begin
  hari:=29;
  end
  else if (bulan=2) and (kabisat=0) then
  begin
  hari:=28;
  end;

//output

  writeln('jumlah hari pada tahun ', tahun, ' dan bulan ', bulan, ' adal

  readln;

end.
```

input

```
Warning: Only one source file supported, changing
pas" into "Soal1_AbiNubliRidhoi_261401056.pas"
Warning: Only one source file supported, changing
_AbiNubliRidhoi_261401056.pas" into "Soal2_AbiNubl
Warning: Only one source file supported, changing
_AbiNubliRidhoi_261401056.pas" into "Soal9_AbiNubl
Free Pascal Compiler version 3.2.2+dfsg-32 [2024/0
Copyright (c) 1993-2021 by Florian Klaempfl and ot
Target OS: Linux for x86-64
Compiling Soal9_AbiNubliRidhoi_261401056.pas
Linking a.out
54 lines compiled, 0.1 sec
3 warning(s) issued
Masukkan tahun: 2008
Masukkan bulan: 2
jumlah hari pada tahun 2008 dan bulan 2 adalah 29
```

SOAL 10



```
//Abi Nubli Ridhoi 261401056 Kom A



program tubes10;
uses crt;
var
hari:integer;

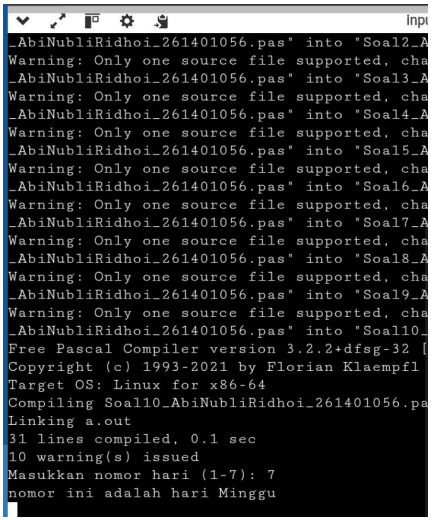
begin
  write('Masukkan nomor hari (1-7): ');readln(hari);

  //menentukan hari dengan nomor dan langsung dioutput

  case hari of
  1:writeln('nomor ini adalah hari Senin');
  2:writeln('nomor ini adalah hari Selasa');
  3:writeln('nomor ini adalah hari Rabu');
  4:writeln('nomor ini adalah hari Kamis');
  5:writeln('nomor ini adalah hari Jumat');
  6:writeln('nomor ini adalah hari Sabtu');
  7:writeln('nomor ini adalah hari Minggu');

  else
  begin
  writeln('Tidak ada dalam pilihan')
  end;
  end;
  readln;

end.
```



```
_AbiNubliRidhoi_261401056.pas" into "Soal2_A
Warning: Only one source file supported, cha
_AbiNubliRidhoi_261401056.pas" into "Soal3_A
Warning: Only one source file supported, cha
_AbiNubliRidhoi_261401056.pas" into "Soal4_A
Warning: Only one source file supported, cha
_AbiNubliRidhoi_261401056.pas" into "Soal5_A
Warning: Only one source file supported, cha
_AbiNubliRidhoi_261401056.pas" into "Soal6_A
Warning: Only one source file supported, cha
_AbiNubliRidhoi_261401056.pas" into "Soal7_A
Warning: Only one source file supported, cha
_AbiNubliRidhoi_261401056.pas" into "Soal8_A
Warning: Only one source file supported, cha
_AbiNubliRidhoi_261401056.pas" into "Soal9_A
Warning: Only one source file supported, cha
_AbiNubliRidhoi_261401056.pas" into "Soal10_
Free Pascal Compiler version 3.2.2+dfsg-32 [
Copyright (c) 1993-2021 by Florian Klaempfl
Target OS: Linux for x86-64
Compiling Soal10_AbiNubliRidhoi_261401056.pa
Linking a.out
31 lines compiled, 0.1 sec
10 warning(s) issued
Masukkan nomor hari (1-7): 7
nomor ini adalah hari Minggu
```